

UML Diagrams

bambu-viewer

Black Sphere Industries

Generated 2026-03-23 | D2 Diagrams

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[IMG] File Load and 3D Rendering

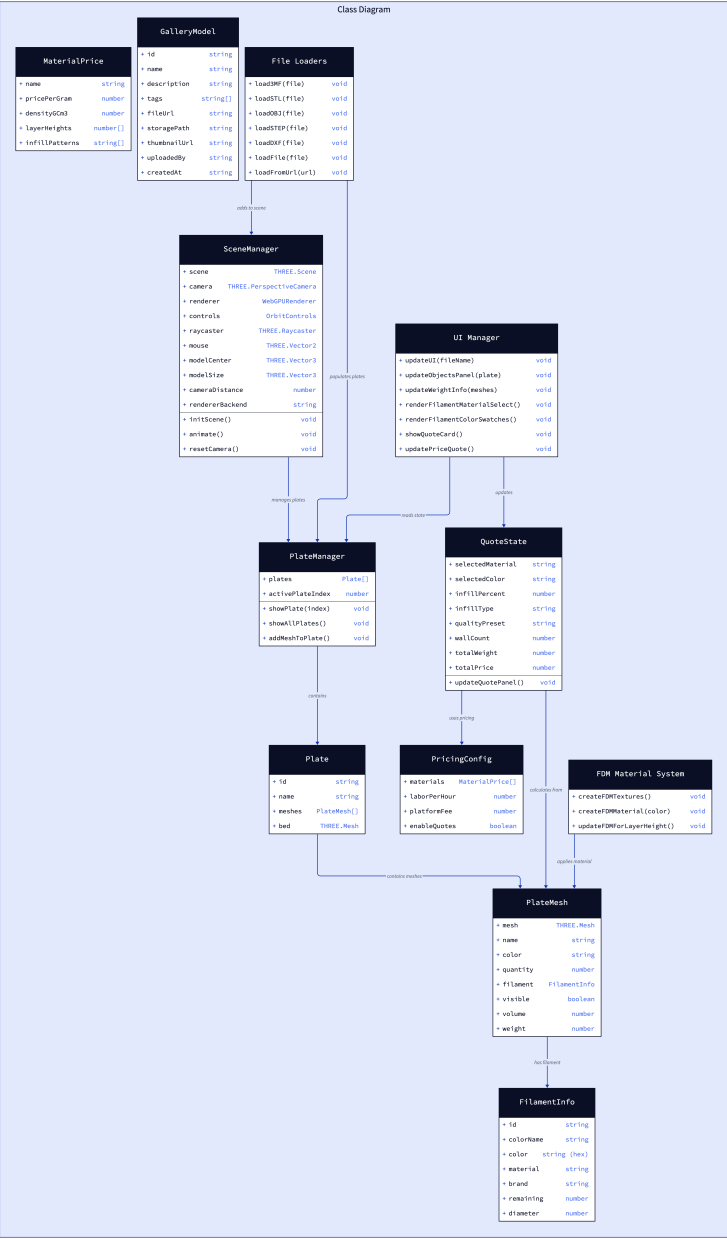
[IMG] Gallery Browse and Quote Flow

[SRC] Embedded Viewer in Shop

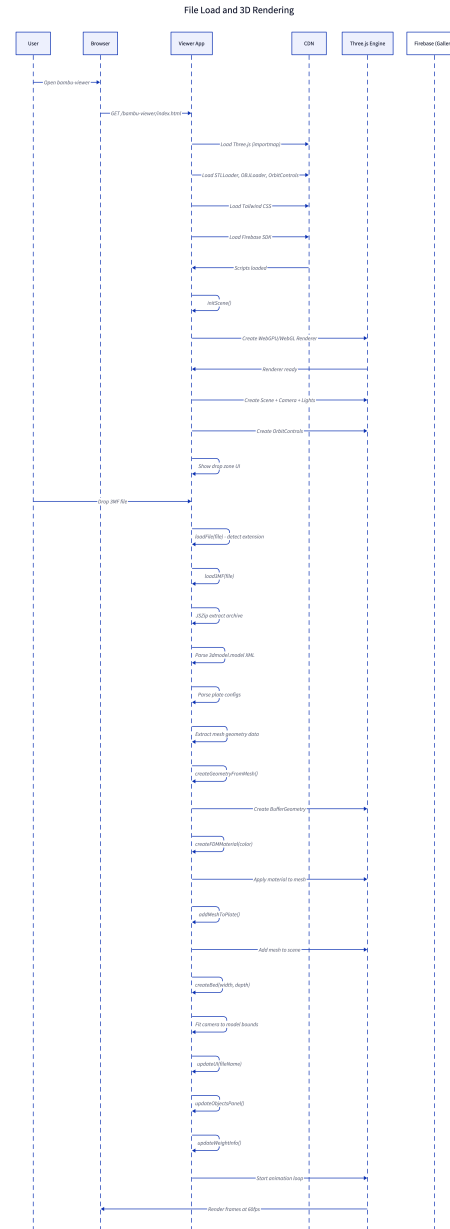
[IMG] Component Architecture

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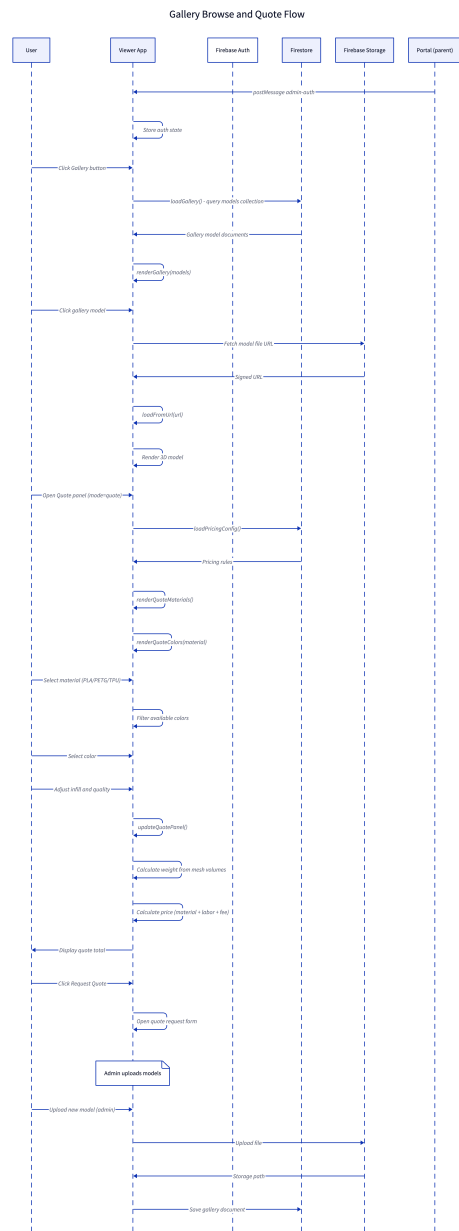
Class Diagram



File Load and 3D Rendering



Gallery Browse and Quote Flow



Embedded Viewer in Shop

```
seq-embed: Embedded Viewer in Shop {
  shape: sequence_diagram

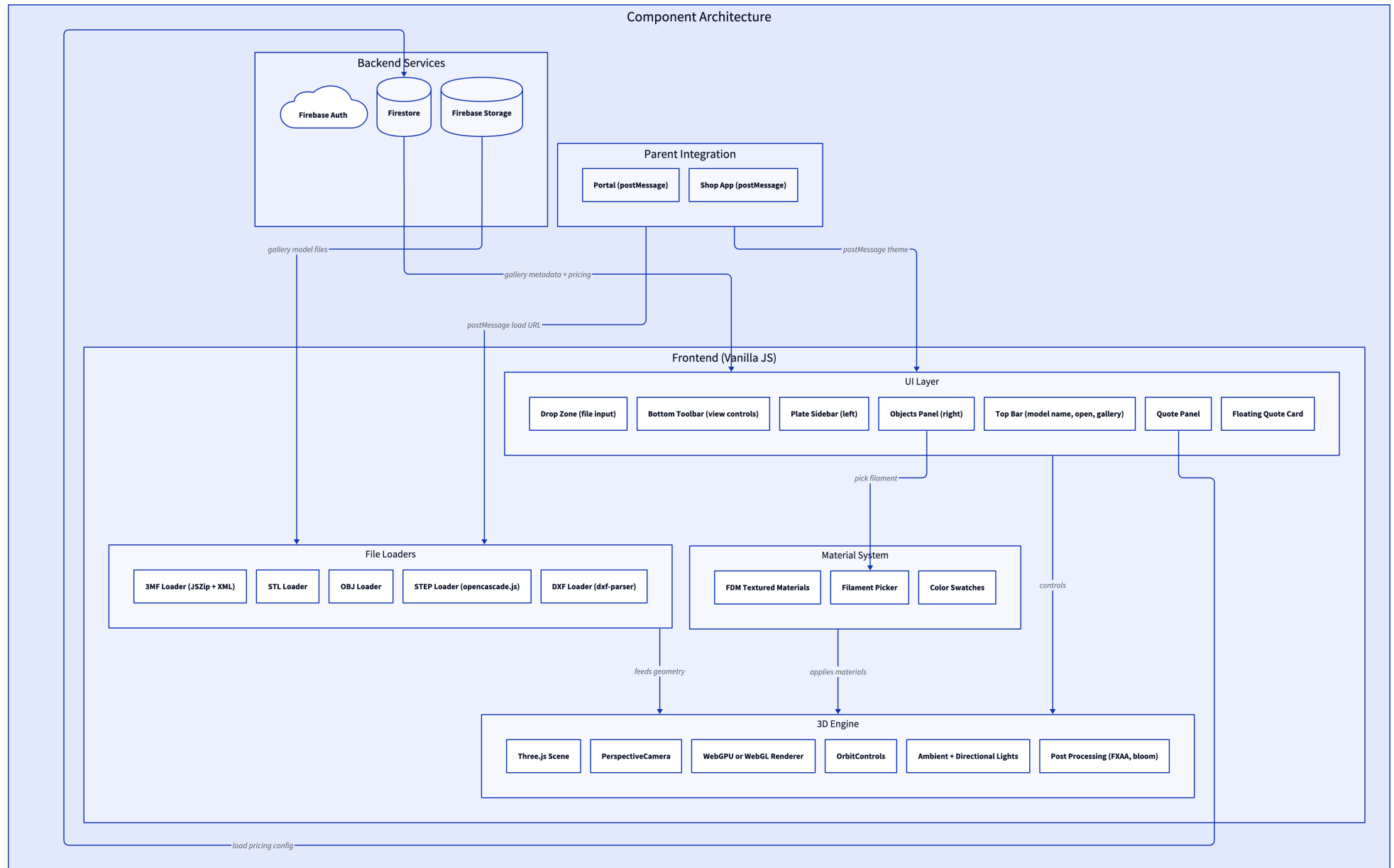
  customer: Customer
  shop: Shop App
  viewer: Bambu Viewer (iframe)
  threejs: Three.js

  customer -> shop: View product with 3D model
  shop -> shop: Check product.default3D or models[]
  shop -> viewer: Embed viewer iframe
  shop -> viewer: postMessage bambu-viewer-load
  shop->>viewer: "load payload": {md
    type: bambu-viewer-load
    url: model STL/3MF URL
  }
  viewer -> viewer: loadFromUrl(url)
  viewer -> threejs: Parse and render model
  threejs -> viewer: Model displayed
  viewer -> customer: Interactive 3D preview

  customer -> shop: Select different variant
  shop -> shop: switchModel(newIndex)
  shop -> viewer: postMessage bambu-viewer-load (new URL)
  viewer -> viewer: Clear scene
  viewer -> viewer: loadFromUrl(newUrl)
  viewer -> customer: Updated 3D preview

  customer -> viewer: Rotate/zoom model
  viewer -> threejs: OrbitControls interaction
  threejs -> viewer: Re-render frame
}
```

Component Architecture



Application States

